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## The essentials

### Introduction and Motivation

**LDA** (Linear Discriminant Analysis) is a **supervised latent analysis** method that makes it possible to find optimal latent factors for discrimination between classes.

## The fundamental problem

Imagine that you work in a postal sorting center. You have letters with 100 different measurements (weight, size, color, texture...). You must sort them automatically by type (letter, parcel, magazine). How can you find **the dimension or dimensions that best separate the categories**?

This is exactly what LDA solves: finding **the axes that maximize separation between classes** while minimizing dispersion within each class.

### Fundamental difference with PCA:

PCA looks at **all** the data and searches for directions of maximum variation. LDA first looks at **the labels** (the classes) and searches for the directions that best separate these classes.

**Warning:** This LDA should not be confused with **Latent Dirichlet Allocation** (used in NLP).

## Supervised learning

Let data  $X \subset \Omega \subseteq \mathbb{R}^D$  be associated with categories (class labels)  $Y = [[M]] \subset \mathbb{N}$ .

**Supervised learning** consists of finding (learning) the parameters  $\theta$  of a learner (function)  $\phi_\theta$  in order to minimize the loss  $L_{X \times Y}(\theta)$  incurred when predicting class labels.

$$\begin{aligned}\phi_\theta : \Omega &\rightarrow Y \\ x_i &\mapsto \phi_\theta(x_i) = \tilde{y}_i\end{aligned}$$

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## Geometric intuition: Separating colored point clouds

Imagine points of two colors (red and blue) scattered on a plane. You can only look along a single direction (like a flashlight that illuminates a line).

**Question:** How should the flashlight be oriented so that, on this line, the red and blue points are **as separated as possible**?

**Essential idea:** LDA looks for the **dividing line**, like a referee drawing a line between two teams. It finds the direction that best separates the groups, while ensuring that each team remains clustered.

We seek a **direction**  $a \in \mathbb{R}^D$  such that the projection of the data onto this direction  $\hat{X} = \text{Proj}_a(X)$  exhibits **optimal properties for discrimination** between classes.

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## The two competing objectives

### Objective 1: Move the centers apart (Between-class)

“Push the means as far apart as possible”

If the class projections are **well separated**, then they can be easily discriminated.

**Objective:** Search for the direction  $a$  where inter-class discrimination is **maximal**.

#### **i** Mathematical details: Between-class formulation

The direction  $a$  must be **collinear** with the line  $\mu_1 - \mu_2$  that connects the centers of mass of the classes, because:

$$|\hat{\mu}_1 - \hat{\mu}_2|^2 = \left| \frac{a^T \mu_1}{|a|^2} a - \frac{a^T \mu_2}{|a|^2} a \right|^2 = \frac{1}{|a|^2} (a^T (\mu_1 - \mu_2))^2$$

This expression is **maximal** when  $a = \lambda(\mu_1 - \mu_2)$ .

**Property:** The center of mass of the projected data is the projection of the center of mass of the original data.

### Objective 2: Group each class (Within-class)

“Keep each class compact”

We also seek to **minimize the variance** of each class when projected individually.

For each class  $k$ , the normalized variance of the projection is:

$$\hat{\sigma}^2 * k^2 = \frac{1}{N_k} \sum y_i^2 = k(\hat{x}_i - \hat{\mu}_k)^T (\hat{x}_i - \hat{\mu}_k)$$

**The dilemma:** Sometimes, pulling the centers apart separates them, but it can also stretch the classes. The right compromise must be found.

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## Fisher criterion: The mathematical compromise

### Formulation for 2 classes

The **Fisher criterion** maximizes the ratio:

$$\arg \max_a \frac{|\hat{\mu}_1 - \hat{\mu}_2|^2}{\hat{\sigma}_1^2 + \hat{\sigma}_2^2}$$

### Interpretation:

- **Numerator:** Distance between projected means (between-class) → to **maximize**
- **Denominator:** Sum of within-class variances → to **minimize**

### Mnemonic trick:

- **B** as in **Between** = **B** for “Separate well” (the centers)
- **W** as in **Within** = “Group together” (each class)

### Generalization to M classes

**i** Mathematical details: Covariance matrices

#### Between-class covariance matrix:

Let  $B = [\mu_1 - \mu, \dots, \mu_M - \mu]$  be the matrix of centered class means, where:

- $\mu = \frac{1}{N} \sum_i x_i$  is the global mean
- $N = \sum_k N_k$  is the total number of points

$$\Sigma_b = \frac{1}{M} B B^T$$

#### Criterion to maximize:

$$\frac{1}{M} \sum_k (\hat{\mu}_k - \hat{\mu})^T (\hat{\mu}_k - \hat{\mu}) = \frac{1}{|a|^2} a^T \Sigma_b a$$

#### Within-class covariance matrix:

Let  $A_k = [x_1 - \mu_k, \dots, x_{N_k} - \mu_k]$  with  $y_i = k$ , the matrix of centered data for class  $k$ .

$$\Sigma_w = \sum_k \frac{1}{N_k} A_k A_k^T$$

#### Criterion to minimize:

$$\sum_k \frac{1}{N_k} \sum_{y_i=k} (\hat{x}_i - \hat{\mu}_k)^T (\hat{x}_i - \hat{\mu}_k) = \frac{1}{a^T a} a^T \Sigma_w a$$

The **generalized Fisher criterion** combines the between-class and within-class criteria:

$$\arg \max_a J(a) = \arg \max_a \frac{a^T \Sigma_b a}{a^T \Sigma_w a}$$

where:

- $\Sigma_b$  = “between-class” covariance matrix → measures separation of centers
- $\Sigma_w$  = “within-class” covariance matrix → measures internal dispersion

## Solution: Eigenvalue problem

### Solution by differentiation

#### **i** Mathematical details: Derivation

To find the maximum, we compute the derivative:

$$\frac{\partial J(a)}{\partial a} = \frac{\Sigma_b a (a^T \Sigma_w a) - \Sigma_w a (a^T \Sigma_b a)}{(a^T \Sigma_w a)^2} = 0$$

**Resulting equation:**

$$\Sigma_b a = J(a) \Sigma_w a$$

Which can be rewritten as:

$$\Sigma_w^{-1} \Sigma_b a = J(a) a$$

**Solution:**  $a$  is a solution of the **generalized eigenvalue system**.

The optimal direction  $a$  is the **first eigenvector** of  $\Sigma_w^{-1} \Sigma_b$  (with eigenvalue  $\lambda_1 = J(a)$ ).

## Dimensional limitation

### Why only M-1 dimensions?

The eigenvectors corresponding to the **largest eigenvalues**  $\lambda_i$  are the most discriminative dimensions:

$$a_1, a_2, \dots, a_p \quad \text{with} \quad \lambda_1 > \lambda_2 > \dots > \lambda_p$$

**Fundamental property:**  $M$  classes can be discriminated in a subspace of dimension **at most**  $(M - 1)$ .

**Consequence:** There are only  $M - 1$  **non-zero** eigenvalues.

**Analogy:** With 3 points (3 classes), you can at most place them in a plane (2D). With 2 points (2 classes), you can at most place them on a line (1D).

#### **i** Mathematical details: Case $M = 2$

For two classes, one can show that:

$$BB^T = (\mu_1 - \mu)(\mu_1 - \mu)^T + (\mu_2 - \mu)(\mu_2 - \mu)^T \propto (\mu_1 - \mu_2)(\mu_1 - \mu_2)^T$$

#### **Implications:**

- $BB^T a$  is a vector in the direction  $(\mu_1 - \mu_2)$
- The solution is  $a = \lambda \Sigma_w^{-1} (\mu_1 - \mu_2)$

**Pitfall to avoid:** If you keep more than  $M-1$  dimensions, the additional directions will have a zero eigenvalue  $\rightarrow$  not discriminative.

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## Practical application of LDA

### Complete LDA algorithm

#### Step 1: Prepare the data

- Check that each class has enough examples (at least  $D+1$  to avoid singular  $\Sigma_w$ )
- Optional: standardize if the scales are very different

#### Step 2: Compute the key matrices

- $\Sigma_w$  (within): average of class covariances
- $\Sigma_b$  (between): covariance of class centers

### Step 3: Solve the eigenvalue problem

Compute the eigenvectors of  $\Sigma_w^{-1}\Sigma_b$

### Step 4: Keep the first M-1 directions

Sort by decreasing eigenvalues, keep the most important ones

### Step 5: Project and classify

- Project the data onto the discriminant subspace
- Classify (for example using a nearest-neighbor rule)

## LDA as a classifier: The Gaussian hypothesis

### Probabilistic modeling

For a new data point  $j$ :

- $x_j$  is **known**
- $y_j$  is **unknown**

**Question:** To which class  $k$  does point  $j$  belong?

We define the class random variable  $C$  and use **Bayes' rule**:

$$\mathbb{P}(C = k|x_j) = \frac{\mathbb{P}(x_j|C = k)\mathbb{P}(C = k)}{\mathbb{P}(x_j)}$$

### Components:

- $\mathbb{P}(x_j|C = k)$  : **likelihood**
- $\mathbb{P}(C = k)$  : **prior**
- $\mathbb{P}(x_j)$  : **evidence** (normalization)

## Gaussian approximation

LDA assumes that **each class follows a Gaussian distribution**:

$$\mathbb{P}(x|C = k) \sim \mathcal{N}(\mu_k, \Sigma_k)$$

with:

- Different means for each class
- **The same covariance matrix** for all classes (crucial!)

**i** Mathematical details: Maximum likelihood

**Approximation:**

$$\mathbb{P}(x|C = k) \propto \exp\left(-\frac{1}{2}(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k)\right)$$

**Uniform prior:**  $\mathbb{P}(C = k) = \frac{1}{M}$  (all classes equally likely)

**Simplification:** The evidence  $\mathbb{P}(x_j)$  is **ignored** because it is constant for all classes (normalization).

**Decision rule:**

$$\delta(x) = \arg \max_k \mathbb{P}(x|C = k)$$

**Interpretation:**  $x$  is assigned to the class  $k$  that maximizes the likelihood.

**Why is the boundary linear?**

With this assumption, the boundary between two classes is **linear** (hence the “L” in LDA).

**Why?** The equal-density contours of two Gaussians with the same covariance are hyper-planes.



## Optimality of LDA

### Optimality conditions

**LDA is optimal** when the  $M$  classes each follow a **Gaussian distribution** with the same covariance matrix.

#### Justification:

The discrimination criteria are based on the covariance matrices  $\Sigma_w$  and  $\Sigma_b$ , which fully characterize Gaussian distributions.

### When LDA works well (and less well)

#### Ideal conditions

- Classes are **approximately Gaussian**
- Class covariances are **similar**
- Enough samples per class (at least  $10\times$  the number of dimensions)
- **Linear** relationships between variables

#### Problematic situations

- **Very different covariances**  $\rightarrow$  use QDA (Quadratic Discriminant Analysis)
- **Non-linear relationships**  $\rightarrow$  Kernel LDA or non-linear methods
- **Few samples** relative to dimensions  $\rightarrow$  regularize or reduce dimensionality first
- **Non-Gaussian classes**  $\rightarrow$  other classifiers (random forests, SVM...)

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## LDA vs QDA: The linear vs quadratic dilemma

### The crucial difference

- **LDA:** Assumes **the same covariance** for all classes  $\rightarrow$  linear boundaries
- **QDA:** Assumes **different covariances**  $\rightarrow$  quadratic boundaries

**Practical rule:** If you have little data, LDA is more robust (fewer parameters to estimate).

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## LDA vs PCA comparison

### Fundamental differences

#### PCA (unsupervised):

- Maximizes **global variance**
- Ignores class labels
- Finds directions of maximum variance
- Based on  $\Sigma$  (total covariance)

#### LDA (supervised):

- Maximizes **class separation**
- Uses class labels
- Finds discriminative directions
- Based on  $\Sigma_w^{-1}\Sigma_b$  (covariance ratio)

### Complementarity

**In practice:** PCA and LDA can be combined:

- PCA for initial dimensionality reduction (noise removal)
- LDA to maximize discrimination

**Concrete example:** Face recognition with Fisherfaces (LDA) as opposed to Eigenfaces (PCA).

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#### Condensed revision sheet

##### In 30 seconds:

- LDA = supervised dimensionality reduction (uses labels)
- Maximizes (distance between centers) / (dispersion within classes)
- Solution = eigenvectors of  $\Sigma_w^{-1}\Sigma_b$
- At most M-1 discriminative dimensions
- Assumes: Gaussian classes, identical covariances
- Useful for: classification, supervised visualization
- Alternative: QDA if covariances differ

## ! Mathematical concepts to master

To fully understand LDA:

### **Linear algebra:**

- Eigenvectors and eigenvalues
- Generalized eigenvalue system
- Covariance matrices
- Orthogonal projection

### **Statistics:**

- Mean and variance
- Covariance and correlation
- Multivariate normal distribution
- Bayes' rule

### **Optimization:**

- Matrix function differentiation
- Constrained optimization
- Rayleigh quotient method